

Foundation (Jalapeno)

Q1. Solve the inequality $2x + 5 > 11$.

- (A) $x > 3$
- (B) $x < 3$
- (C) $x > -3$
- (D) $x < -3$
- (E) $x = 3$

Q2. Solve the inequality $x - 2 < 7$.

- (A) $x < 5$
- (B) $x > 5$
- (C) $x = 5$
- (D) $x < -5$
- (E) $x > -5$

Q3. Graph the solution to $x > 2$ on a number line.

- (A) A line to the left of 2
- (B) A line to the right of 2
- (C) A point at 2
- (D) Two points at 2 and 3
- (E) A circle around 2

Q4. Write the solution to $x < 4$ in interval notation.

- (A) $(-\infty, 4)$
- (B) $(-\infty, 4]$
- (C) $[4, \infty)$
- (D) $(-4, \infty)$
- (E) $(-4, 4)$

Q5. Solve the inequality $3x - 2 < 10$.

- (A) $x < 4$
- (B) $x > 4$
- (C) $x = 4$
- (D) $x < -4$
- (E) $x > -4$

Q6. Graph the solution to $x + 1 > 3$ on a number line.

- (A) A line to the left of 2
- (B) A line to the right of 2
- (C) A point at 2
- (D) Two points at 2 and 3
- (E) A circle around 2

Q7. Write the solution to $2x > 6$ in interval notation.

- (A) $(-\infty, 3)$
- (B) $(-\infty, 3]$
- (C) $[3, \infty)$
- (D) $(3, \infty)$
- (E) $(-3, 3)$

Q8. Solve the inequality $x - 4 \leq 2$.

- (A) $x \leq 6$
- (B) $x \geq 6$
- (C) $x = 6$
- (D) $x < 6$
- (E) $x > 6$

Open Ended Questions (FRQ)

Q9. Solve the inequality $2x + 5 > 3x - 2$ and write the solution in interval notation.

Q10. Graph the solution to $x + 2 > 5$ on a number line and write the solution in interval notation.

Chef's Tips

- Check for any misconceptions about inequality signs
- Ensure students are using the correct operations to solve inequalities
- Remind students to consider the direction of the inequality when graphing